

Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects*

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Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

Keywords: difference-in-differences, two-way fixed effects, pretrend test

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following differences. First, in the setting where there is no never-treated group, our method uses the last cohort to be treated as a control group, whereas Callaway and Sant’Anna (2020a) use the set of not-yet-treated cohorts. Our method and theirs thus rely on different, but non-nested parallel trends assumptions. Second, our estimation method can be cast as a regression specification and thus may be more familiar to applied researchers. However, a third difference is that the procedure of Callaway and Sant’Anna (2020a) allows for conditioning on time-varying covariates. de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018) respectively propose alternative estimators and diagnostic tools for estimation of causal effects in staggered settings, but do not consider the estimation of the dynamic path of treatment effects as we do.

2 Event studies design

In this section we first formalize the “event studies design”. As discussed in Section 2.3, based on how this term is deployed in the empirical literature, an *event study* design is a *staggered adoption* design where units are treated at different times, and there may or may not be never treated units. It also nests a *difference-in-differences* design, where units are either first treated at time t_0 or never treated.

Specifically, we consider a setting with a random sample of N units observed over $T + 1$ time periods, where T is fixed. For each $i \in \{0, \dots, N\}$ and $t \in \{0, \dots, T\}$, we observe the outcome $Y_{i,t}$ and treatment status $D_{i,t} \in \{0, 1\}$: $D_{i,t} = 1$ if i is treated in period t and $D_{i,t} = 0$ if i is not treated in period t . Throughout we assume that the observations $\{Y_{i,t}, D_{i,t}\}_{t=0}^T$ are independent and identically distributed (i.i.d.).

In the general case of event studies we focus on an *absorbing* treatment such that the treatment status over time is a non-decreasing sequence of zeros and then ones, i.e. $D_{i,s} \leq D_{i,t}$ for $s < t$. We can thus uniquely characterize a treatment path by the time period of the initial treatment, denoted with $E_i = \min \{t : D_{i,t} = 1\}$. If unit i is never treated i.e. $D_{i,t} = 0$ for all t , we set $E_i = \infty$. Based on when they first receive the treatment, we can also uniquely categorize units into disjoint cohorts e for $e \in \{0, \dots, T, \infty\}$, where units in cohort e are first treated at the same time $\{i : E_i = e\}$.

We define $Y_{i,t}^e$ to be the potential outcome in period t when unit i is first treated in time period e . We define $Y_{i,t}^\infty$ to be the potential outcome if unit i never receives the treatment, which we call the “baseline outcome”. Since the timing of the initial treatment uniquely characterizes one’s treatment path, we can

for all ℓ and all t . We can represent the relative period bin indicator as

$$\mathbf{1}\{t - E_i \in g\} = \sum_{\ell \in g} \mathbf{1}\{t - E_i = \ell\} = \sum_{\ell \in g} D_{i,t}^\ell. \quad (5)$$

Static specification. For a “static” specification $\mathcal{G}$ contains a single element equal to $g = [0, T]$. The indicator $\mathbf{1}\{t - E_i \in g\}$ is equivalent to an indicator for whether unit i has received its initial treatment by t : $\mathbf{1}\{E_i \leq t\}$. The “static” specification thus takes the following form

$$Y_{i,t} = \alpha_i + \lambda_t + \mu_g \sum_{\ell \geq 0} D_{i,t}^\ell + v_{i,t} \quad (6)$$

and the corresponding set of excluded relative periods is $g^{excl} = [-T, -1]$.

Dynamic specification. “Dynamic” specifications encompass any specifications of $\mathcal{G}$ where $\mathcal{G}$ contains more than one element, thus treatment effects are allowed to vary over time non-parametrically. In its most flexible form, a “fully dynamic” specification takes the following form

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\ell=-K}^{-2} \mu_\ell D_{i,t}^\ell + \sum_{\ell=0}^L \mu_\ell D_{i,t}^\ell + v_{i,t} \quad (7)$$

and the corresponding set of excluded relative periods is $g^{excl} = \{-T, \dots, -K-1, -1, L+1, \dots, T\}$.

Excluding some relative periods from the “fully dynamic” specification is necessary to avoid multi-collinearity, either among the relative period indicators $D_{i,t}^\ell$, or with the unit and time fixed effects. For example, when there are no never-treated units i.e. $\infty \notin \text{supp}(E_i)$ but with a panel balanced in calendar time, we need to exclude at least two relative period indicators in $\mathcal{G}$. These collinearities are discussed by Borusyak and Jaravel (2017): one multi-collinearity comes from the relative period indicators summing to one for every unit $\sum_{\ell \in [-T, T]} D_{i,t}^\ell = 1$, and the other multi-collinearity comes from the linear relationship between two-way fixed effects and the relative period indicators, namely $t - E_i = \ell$.

Excluding relative periods close to the initial treatment is common in practice. Normalizing relative to the period prior to treatment is the most common - six out of the eight papers we survey do so, as reflected in the above specification where we drop $D_{i,t}^{-1}$. The remaining two papers exclude $D_{i,t}^0$.

Excluding distant relative periods is however less common (only one of the eight papers we survey does so). Instead researchers “bin” or “trim” distant relative periods. For “binning”, researchers bin distant

each $CATT_{e,\ell}$ using an interacted two-way fixed effects regression, then estimate the weight $Pr\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ using their sample analogs. In the final step, we average over the cohort-specific estimates associated with relative period ℓ . This method has a similar flavor as the method proposed by Gibbons et al. (2019). They first use an interacted model to estimate the treatment effect for each fixed effect group; the resulting group-specific estimates are averaged to provide the ATE. Their method improves fixed effects regressions in a cross-sectional setting, and our method builds on theirs by improving two-way fixed effects regressions in a panel setting. We therefore follow their terminology in calling our alternative estimator an “interaction-weighted” estimator.

4.1 Interaction-weighted estimator

We describe the estimation procedure in three steps (with more detailed definitions stated in Definition 4 of Online Appendix B).

Step 1. We estimate $CATT_{e,\ell}$ using a linear two-way fixed effects specification that interacts relative period indicators with cohort indicators, excluding indicators for cohorts from some set C :

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{e \notin C} \sum_{\ell \neq -1} \delta_{e,\ell} (\mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell) + \epsilon_{i,t}. \quad (26)$$

The exact specification depends on the cohort shares for a given application. If there is a never-treated cohort, i.e. $\infty \in \text{supp}\{E_i\}$, then we may set $C = \{\infty\}$ and estimate regression (26) on all observations. If there are no never-treated units, i.e. $\infty \notin \text{supp}\{E_i\}$, then we may set $C = \{\max\{E_i\}\}$, i.e. the latest-treated cohort and estimate regression (26) on observations from $t = 0, \dots, \max\{E_i\} - 1$. Lastly, if there is a cohort that is always treated, i.e. $0 \in \text{supp}\{E_i\}$, then we need to exclude this cohort from estimation.

The coefficient estimator $\widehat{\delta}_{e,\ell}$ from regression (26) is a DID estimator for $CATT_{e,\ell}$ with particular choices of pre-periods and control cohorts. As DID is likely a familiar estimator for applied researchers, we separate the more in-depth discussion of its definition, choices of pre-periods, and choices of control cohorts in Section 4.2.

Step 2. We estimate the weights $Pr\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ by sample shares of each cohort in the relevant period(s) $\ell \in g$.

Step 3. To form our IW estimator, we take a weighted average of estimates for $CATT_{e,\ell}$ from Step 1

estimator $\widehat{\delta}_{e,\ell}$ is a DID estimator for $CATT_{e,\ell}$ with pre-period $s = e - 1$ (because we exclude relative period $\ell = -1$) and some choice of control cohorts C . If there is a never-treated cohort i.e. $\infty \in \text{supp}\{E_i\}$, then we set the control cohort to be never-treated units $C = \{\infty\}$. If there are no never-treated units, i.e. $\infty \notin \text{supp}\{E_i\}$, then we set the control cohort $C = \{\max\{E_i\}\}$, i.e. the latest-treated cohort. Among all possible pre-periods, for regression (26) from step 1 of our proposed method we choose $s = e - 1$ and $C = \{\infty\}$ with never-treated units (or $\{\max\{E_i\}\}$ without never-treated units) because the resulting specification is a natural extension of the common specifications of two-way fixed effects regression.

Note that without never-treated units we need to drop time periods $t \geq \max\{E_i\}$ from estimating regression (26) because every unit will be treated in these periods. The DID estimators for $CATT_{e,\ell}$ for $e + \ell \geq \max\{E_i\}$ are thus not well-defined as the control cohort is empty $C = \emptyset$. For example, when there are just two cohorts with $E_i \in \{1, T\}$, one treated at $t = 1$ and the other treated in the last period, we then need to omit interaction terms involving the latest-treated cohorts as well as dropping observations from the last period $t = T$ from estimation.

Under some assumptions, this DID estimator $\widehat{\delta}_{e,\ell}$ is an unbiased and consistent estimator for $CATT_{e,\ell}$, a fact that we build on in deriving the probability limit of the IW estimator. We state this in the following proposition.

Proposition 5. *If Assumptions 1 and 2 hold, then the DID estimator using any pre-period $s < e$ and non-empty control cohorts C is an unbiased and consistent estimator for $CATT_{e,\ell}$.*

It is possible to relax the parallel trends assumption to allow the timing of treatment to depend on covariates. One can estimate $CATT_{e,\ell}$ consistently based on the inverse propensity score reweighted estimator proposed by Abadie (2005) and Callaway and Sant’Anna (2020a), the outcome regression approach by Heckman et al. (1997), and the doubly robust estimator recently proposed by Sant’Anna and Zhao (2020). The resulting estimates $\widehat{\delta}_{e,\ell}$ can then be plugged into step 3 to form our IW estimator. In particular, without covariates and for the case with never treated units, our approach coincides with Callaway and Sant’Anna (2020a). Therefore one can use the did R package developed by Callaway and Sant’Anna (2020b) to form the IW estimator.

The choice of the pre-period s and the control cohorts C depends on the trade-off between relaxing Assumption 1 or Assumption 2. If Assumption 2 is likely to hold, we can choose any pre-period $s < e$ and include not-yet-treated units in control cohorts $C = \{c : c > e + \ell\}$ as in Callaway and Sant’Anna (2020a). This